

RISE IN × STELLAR PRO HACKATHON 2026 · SCALE TRACK

RewardRail

Every verified action pays everyone.
Instantly.

A player finishes a game. Three parties are paid in seconds, the advertiser can audit every cent, and a fraudulent reward can still be taken back.

Stellar testnet

Soroban · Clawback · SEP-6

Soroswap



ONE \$4.00 INSTALL

Three parties, three different clocks

Player

\$1.20

held back until a minimum is reached

Weeks

most never clear it

Publisher

\$1.80

monthly reconciliation

Next cycle

on the platform's numbers

Platform

\$1.00

the same cycle

Next cycle

and it eats the fraud loss

A reward of a few cents costs more to send than it is worth on a consumer payment rail, so a minimum appears. The advertiser cannot verify the invoice it receives. And fraud found after the money has left is written off, because nothing brings it back.

Figures from a mid-size operator running this model across half a dozen markets

BUILT AROUND THE HARD PARTS

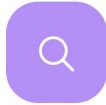
Three things the existing rail cannot do



No threshold

A Stellar transaction costs about 0.00001 XLM, so a reward of a few cents is payable the moment it is earned.

1.20 paid per action



Auditable spend

The budget sits in a Soroban escrow. Every release is a transaction the advertiser reads for itself.

every leg on chain



Reversible rewards

Fraud found after payout is clawed back on the ledger, not written off as a loss.

60s window in the demo

RewardRail sits underneath a rewarded-ads platform. Campaign selection, matching and fraud detection stay where they are. What changes is how a verified action turns into money.

FROM COMPLETED ACTION TO SETTLED PAYOUT

How one verified action becomes money



Advertiser funds

In any asset it holds.
Soroswap routes it to the
campaign currency.

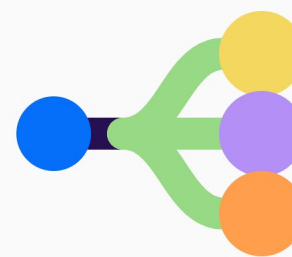
12.00 USDC locked



Validator verifies

Your own service signs a
proof the contract checks
on chain.

paid at most once



Three shares open

Player, publisher
and platform, at
ratios fixed when the
campaign opened.

$1.20 \cdot 1.80 \cdot 1.00$



Fraud reverses

Inside the window the
reward returns to the
budget, on the ledger.

1.20 recovered

The chain never knows whether someone really played. It guarantees the rest: the budget cannot move without a valid proof, ratios cannot deviate, one action is paid at most once, and unspent budget goes back.

WHY STELLAR

Each problem maps to a protocol feature

Problem	Stellar feature	Why it is hard elsewhere
Payout threshold	Fee ~0.00001 XLM	Elsewhere the fee exceeds the reward
Reversing fraud	Clawback, CAP-35	Needs custody or bespoke contract logic
Wallet setup	Sponsored reserves	User must buy a token before existing
Fees for a new user	Fee-bump	Someone has to hold gas
Local currency exit	SEP-6 · SEP-10 anchors	One integration per market, per rail
Conditional budget	Soroban escrow	—

Not a chain chosen and then justified. Remove any one of these and the product stops working.

THE STRONGEST CLAIM

The clawback window is enforced by the ledger

A reward is paid and frozen in one transaction. The player sees the balance and cannot move it — not to a second account, not to an exchange. Without the freeze the window is only a promise: forward the reward, convert it elsewhere, and by the time fraud surfaces the account is empty.

```
npm run prove-auth-lock — the bypass attack, run against testnet  
player holds 1.0000000 REWARD, trustline frozen  
transfer rejected — op_src_not_authorized  
second account received nothing  
clawback still reached the frozen reward
```

Trusted accounts never see the window. It applies where fraud concentrates: new accounts, and only until it closes.

THE LAST MILE

Play a game, get paid in Turkish lira

One run, end to end

1.20 USDC from escrow

SEP-10 with the player's own key

SEP-6 withdrawal opened

rate locked 48.541152 TRY/USDC

USDC sent on chain with memo

Paid out

58.24 TRY

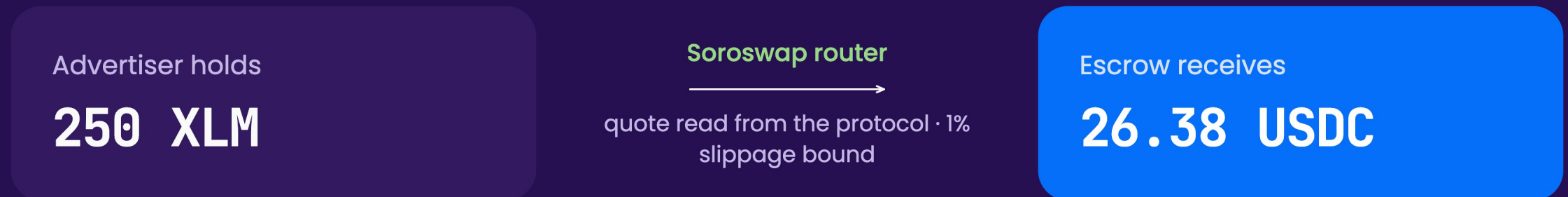
status: completed • fee 0.30

Endpoints discovered from the anchor's **stellar.toml**, the challenge verified against the signing key it publishes. Real: the authentication, the withdrawal, the rate, the payment on chain. Simulated: the bank. Production is a licensed anchor at a different home domain, and the same code.

ECOSYSTEM INTEGRATION

An advertiser funds in whatever it holds

The escrow settles in one currency, because ratios and reserves have to be denominated in something. Left alone, that mismatch becomes the advertiser's problem: acquire the exact asset first, then come back and open a campaign.



A real swap against a pool nobody on this team provided. The aggregator API is mainnet-only and key-gated, so we call the router contract directly, at the address Soroswap publishes for testnet.

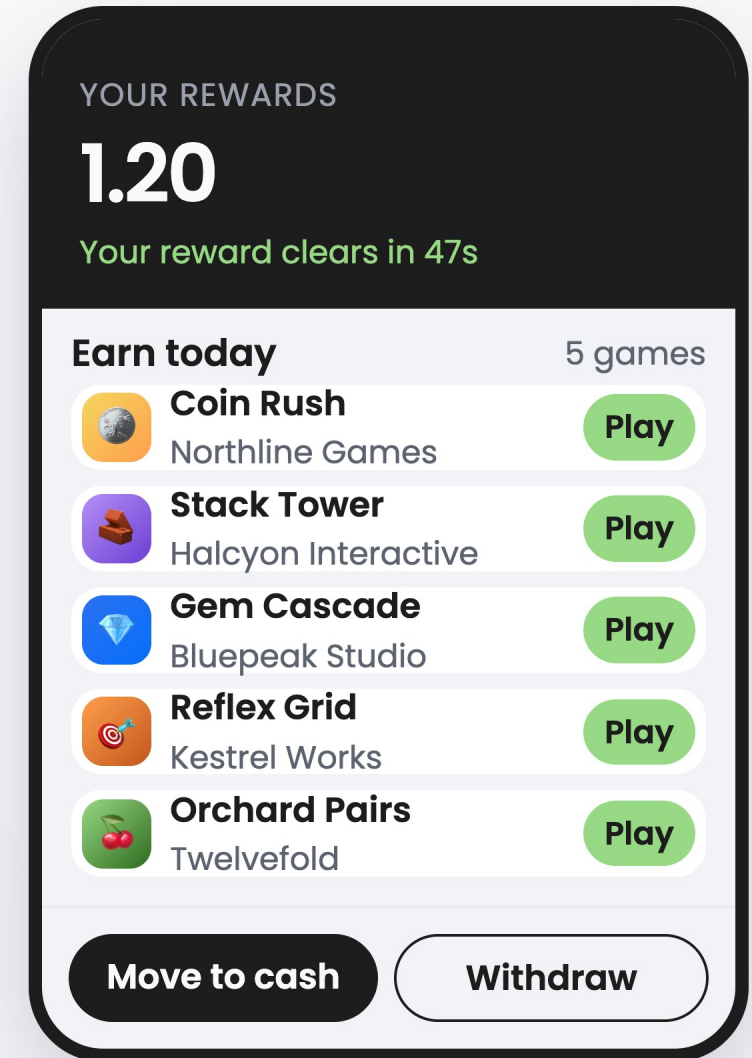
WHAT THE PLAYER HOLDS

The player never learns what any of this is

Sign in with an email. The account is opened with a sponsored reserve and every transaction is fee-bumped, so there is nothing to fund, nothing to install and nothing to keep safe.

Five playable games, a balance, and a withdraw button. The words wallet, seed, gas and blockchain appear nowhere on the screen.

A rewarded-ads user is not a crypto user. Any flow that asks for wallet setup loses them at step one.



Two settled actions, seen from four sides

campaign 8 • 12.00 USDC • 4.00 per action

Advertiser

CAMPAIGN OWNER

Locked
12.00

Released
8.00

Returned
5.20

Campaign opened

02aa98ea...8812b6

Player

FINISHED A GAME

Earned
1.20

Frozen for
60s

Minimum
None

Reward paid

3759c66a...542fa2

Publisher

TRAFFIC SOURCE

Accrued
3.60

Per action
1.80

Minimum
None

Withdrawn on demand

1febf69e...037441

Platform

RUNS THE RISK CALL

Earned
2.00

Reversed
1.20

Written off
None

Clawed back to the budget

9b0b4bb0...697cbe

12.00 in, 8.00 released, 1.20 reversed back into the budget, 5.20 returned at close. Every figure is a balance the contract holds; every hash opens in Stellar Expert.

PROVEN ON TESTNET · SETTLED ON CHAIN

Deployed, and checked rather than claimed

One recorded
run, verified after
the fact

13

hashes re-read
from Horizon

69

tests, contract and
interface

138s

a full rehearsal, most
of it the freeze

3

assumptions proved
by runnable scripts

Escrow contract

CD6HZHGUURVSRWZA0DFF
LC7JX5WCXZCXHEX0FPXA
E5V5ULAD3NDCVTYI

Payout asset — Circle testnet USDC

CBIELTK6YBZJU5UP2WWQ
EUCYKLP6AUNZ2BQ4WWF
EIE3USCIHMXQDAMA

Soroswap router

CCJUD55AG6W5HAI5LRVN
KAE5WDP5XGZBUDS5WNTI
VDU70264UZZE7BRD

npm run prove-clawback · prove-auth-lock · e2e · rehearse

THE WALKTHROUGH

Four minutes, nine steps, every one verifiable

Minute	Step	Proof on screen
0:35	Campaign opening	Soroswap route, escrow balance
1:05	Player signs in by email	Sponsored account, zero XLM
1:30	Plays a game	Settle and reward, two hashes
1:50	Publisher withdraws	Transfer from escrow, no minimum
2:15	Operator flags a fraud	Clawback; honest player untouched
2:45	The reward unfreezes	Burn and redeem, paid from escrow
3:10	Cash out to lira	SEP-6 id, rate and TRY amount

The publisher and fraud steps fill the 60s freeze, so the window is shown by what happens during it.

TRUST BY ARCHITECTURE

What is real, and what is not

✓ Real

The contract and its tests. Clawback and the freeze. Sponsored accounts and fee-bumps. The Soroswap swap, against liquidity we did not provide. The anchor authentication, the withdrawal and the rate. Circle's own testnet USDC.

⚠ Not real

The bank. The anchor simulates the lira leg — no IBAN receives money, no KYC is performed. Player keys are custodial and held by the backend. The tier rule is enforced off chain.

Each of those is written down in the repo beside the code it describes, with the reason it was chosen. Production means a licensed anchor and a KMS — the same protocol against a different counterparty.

What comes after the weekend

01

Close the on-ramp

The lira exit works. Funding a campaign by bank transfer is written and waiting on the anchor's deposit leg.

02

Atomic settlement

Hand the reward asset's SAC admin to the escrow so a payout is one transaction. The documented pattern, in production today as USDT0.

03

Pilot with one operator

A single platform running real campaigns is worth more than breadth, and is the path into SCF and InstAward.

SETTLEMENT YOU CAN INSPECT

**A reward that arrives instantly,
can be audited by the
advertiser, and can still be
taken back when it was
earned by cheating.**

All three on testnet today, and a player paid in Turkish lira at the end of it.

Rise In × Stellar Pro Hackathon 2026 · Scale Track



Open it now

`rewardrail.vercel.app`